

# QUESTION 1

## Volume of a Solid Made by Joining a Cylinder and a Hemisphere

### 1. Understanding the Problem

Write a C program to find the volume of a solid made by joining a cylinder and a hemisphere when their common radius and cylinder height are given.

**Input:** Radius of the cylinder and hemisphere (r), and height of the cylinder (h).

**Output:** Volume of the complete solid.

**Formula:**

Volume of cylinder =  $\pi r^2 h$

Volume of hemisphere =  $(2/3)\pi r^3$

**Total Volume** =  $\pi r^2 h + (2/3)\pi r^3$ , where  $\pi = 3.14159$ .

### 2. Test Case Table

| Test Case | Input(s)      | Expected Output(s) | Actual Output(s) | Result |
|-----------|---------------|--------------------|------------------|--------|
| 1         | r = 3, h = 5  | 169.65             | 169.65           | Pass   |
| 2         | r = 2, h = 4  | 67.02              | 67.02            | Pass   |
| 3         | r = 5, h = 10 | 1047.20            | 1047.20          | Pass   |

### 3. Algorithm

1. Start.
2. Declare variables r, h, and volume.
3. Read the radius r.
4. Read the cylinder height h.
5. Calculate the volume of the cylinder.
6. Calculate the volume of the hemisphere.
7. Add both volumes to obtain the total volume.
8. Display the total volume.
9. Stop.

Evaluation: For r = 3 and h = 5, the calculated volume is approximately 169.65, matching the expected result.

### 4. Flowchart

START → Input r and h → Calculate  $V_c = \pi r^2 h$  → Calculate  $V_h = (2/3)\pi r^3$  → Calculate  $V = V_c + V_h$  → Display V → STOP

Use an oval for START/STOP, parallelograms for input/output, rectangles for calculations, and arrows to connect the steps.

### 5. C Program

```
#include <stdio.h>

int main()
{
```